QVLA: NOT ALL CHANNELS ARE EQUAL IN VISIONLANGUAGE-ACTION MODEL'S QUANTIZATION

26/04/03

[기존 문제]

1. 현재 VLA 모델의 크기

The rapid evolution of foundation models (Touvron et al., 2023a;b) has significantly advanced embodied intelligence, empowering Vision-Language-Action (VLA) models like OpenVLA (Kim et al., 2024) to synthesize complex, executable actions from visual inputs and linguistic directives. These models enhance cross-task generalization and semantic reasoning, elevating the efficacy of robotic manipulation. However, their immense computational and memory demands, which often exceed 14 GB in standard half-precision for a 7B model, present a critical barrier to deployment. For instance, running such a model on a widely-used robotic platform like the NVIDIA Jetson AGX Orin can result in inference latencies of several hundred milliseconds per action, far too slow for the fluid, real-time control required in dynamic environments. This performance gap necessitates aggressive compression methodologies, such as pruning and quantization, to achieve practical inference speeds while sustaining precise control (Yang et al., 2025; Song et al., 2025b; Wen et al., 2024;

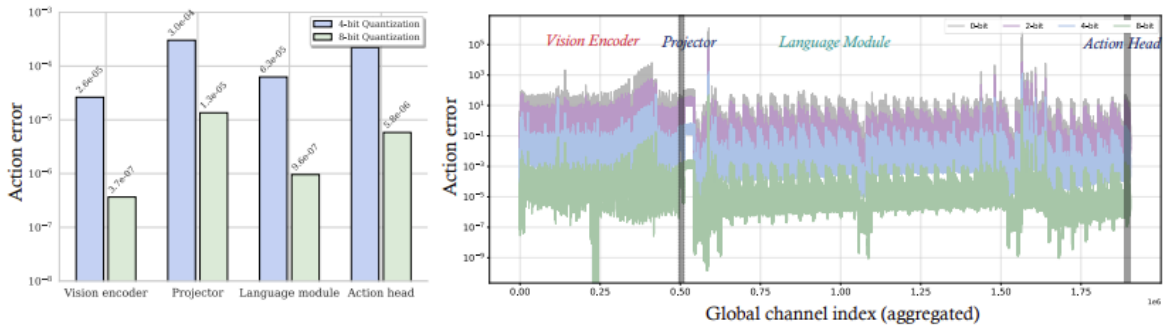
VLA 모델의 크기로 인해, 로봇에 올리지 못하고 클라우드 환경에서 쓸 수밖에 없는 상태. 이로 인해 네트워크 latency 등 추론 속도와 memory 등 현실적 제약 존재.

⇨ **VLA 모델 경량화의 필요성**

2. VLA Quantization

VLA는 action이 누적됨에 따라 최종 task 성공에 대한 오차가 증폭.

또한, VLA는 **LLM decoder(1), vision encoder(2), multi-modal projection layer(3), action decoder(4)** 총 4개의 모듈로 구성됨에 따라, 기존 LLM/MLLM에 썼던 quantization 기법을 적용했을 때 잘 되지 않았음



(a) Quantization Results Across Modules (b) Quantization Results Across Channels

⇨ 모듈 별로, 나아가 channel 별로 action 출력에 대한 민감도가 전부 다름으로 이를 고려하여, **channel-wise quantization**을 적용하자.

[Related Work]

Quantization methods. Model quantization is a cornerstone technique for efficient deep learning deployment, reducing memory footprint and computational latency by representing weights and activations with low-bit integers. The central challenge lies in minimizing the ensuing accuracy degradation. Two primary methodologies dominate the field, *i.e.,* Post-Training Quantization (PTQ) (Frantar et al., 2022; Li et al., 2021; Nagel et al., 2020) and Quantization-Aware Training (QAT) (Jacob et al., 2018; Esser et al., 2020). PTQ offers a low-cost solution by quantizing a pretrained model with a small calibration set and no retraining. In contrast, QAT simulates the effects of quantization during training, allowing the model to adapt its parameters to mitigate precision loss. Recent advances have focused on mitigating the challenges of quantizing large models, particularly the presence of outliers and activation-induced quantization difficulties. Methods like SmoothQuant (Xiao et al., 2022) re-scale weights and activations to create a more favorable quantization landscape, while AWQ (Lin et al., 2023) preserves salient weights and OmniQuant (Shao et al., 2024) smooths weight and activation outliers. To handle pernicious outliers in weights and activations, another research thrust employs learned rotations or permutations to redistribute value distributions, exemplified by methods like QuaRot (Ashkboos et al., 2024) and SpinQuant (Liu et al., 2024c), enabling robust 4-bit quantization. However, a common limitation across these methods is the assumption of uniform precision, where a single bit-width is applied globally or, at best, per layer (*e.g.*, HAWQ (Dong et al., 2019; 2020)). This coarse-grained approach is fundamentally misaligned with the requirements of embodied policies. VLA models exhibit heterogeneous sensitivity across their architecture. For instance, subtle shifts in action-generation layers can lead to catastrophic failures in control, a fragility not typically observed in standard perception or language tasks. Motivated by this critical shortcoming, we propose a fine-grained, action-guided, channel-wise quantization framework.

기존 LLM method들은 모델 전체나 특정 layer 단위로, 동일한 bit 수를 일괄 적용하는 "uniform precision"을 가정

하지만, VLA에서 "Action"에 미치는 민감도를 channel 별로 세밀하게 bit를 할당해야 함

[About VLA]

3.1 PRELIMINARIES

Vision-Language-Action (VLA) models. A VLA model governs the behavior of an embodied agent by defining a policy, denoted as Π, that maps high-dimensional sensory inputs and a language directive to a sequence of actions. The model operates within a sequential decision-making framework. Formally, at each discrete timestep t, the agent perceives its environment through visual observations $\mathcal{V}_t$ (*e.g.*, RGB images) and receives a time-invariant language instruction (prompt) p. The policy's role is to predict an action sequence $\mathcal{A}_t$ (*e.g.*, end-effector poses or joint velocities),

$$\Pi_\theta(\mathcal{A}_t|\mathcal{V}_t, p). \tag{1}$$

The parameter set θ typically comprises: ♠ vision encoders θ_{vis} that processes the high-dimensional visual inputs $\mathcal{V}_t$ into compact feature representations; ♠ a projection layer θ_{proj} that maps these visual features into the multimodal embedding space shared with the language modality; ♠ a Large

간단하게 말해서, 현재 scene에 대한 visual observation과 지시명령 prompt가 주어지면, 그 상황에 맞는 Action을 내뱉는 것이 VLA.

이때 각 timestep별로 sequential하게 action을 수행하고 관측하고 이를 반복.

[Sensitivity Analysis]

1. 모듈 별 sensitivity

   (Action decoder/Multi-modal projection)>Language Model>Vision Encoder

   순으로 quantization에 따른 degradation에 민감함

   이에 action decoder는 full precision을 유지하면서, vision encoder와 language model을 중점적으로 quantization 진행.

components serve as the final nexus for translating multimodal understanding into physical action. Any perturbations introduced here propagate directly and without mitigation to the output action distribution, leading to significant and often erroneous deviations in the agent's behavior. These findings underscore that an adaptive quantization strategy is not merely beneficial but essential for preserving the functional integrity of VLA models at low bit-widths.

2. Channel 별 sensitivity

   Per channel 별, mixed-precision strategy 사용

   이때, {16, 8, 4, 2, 0(pruning)} bit 각 조합에 대해 sensitivity check.

   Sensitivity는 다음과 같이 각 time step 별과, 전체 episode 누적에 대한 sensitivity을 구할 수 있음

   Sensitivity는 action space에 대한 변화를 기준으로 함.

To effectively identify these salient channels, we propose a novel sensitivity metric grounded directly in the action space rather than the intermediate feature space. More specifically, for a given layer l and a specific channel c within it, we isolate its impact by quantizing only that channel to a bit-width $b \in \{0, 2, 4, 8, 16\}$, while all other parameters remain in full precision. We then define the single-step action sensitivity as the expected squared L2 norm of the resulting action deviation,

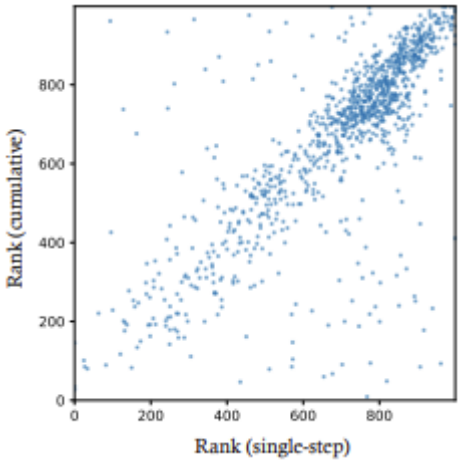
$$s_{l,c}^{(b)} = \mathbb{E}_{x \sim \mathcal{D}} \left[\left\| \tilde{\mathcal{A}}_{l,c}^{(b)}(\mathcal{V}, l) - \mathcal{A}^*(\mathcal{V}, l) \right\|_2^2 \right], \tag{4}$$

where $\tilde{\mathcal{A}}_{l,c}^{(b)}(x)$ is the perturbed action generated with the quantized channel, and $\mathcal{A}^*(x)$ is the reference action from the full-precision model. However, single-step error metrics like $s_{l,c}^{(b)}$ may not fully capture the error accumulation that occurs in long-horizon, autoregressive tasks. A small, initial error can compound over a sequence of actions. To account for this, we introduce a cumulative sensitivity metric, which measures the total deviation over an entire episode:

$$S_{l,c}^{(b)} = \mathbb{E} \left[\sum_{t=1}^{T} \left\| \tilde{\mathcal{A}}_{l,c}^{(b)}(\mathcal{V}_t, l) - \mathcal{A}^*(\mathcal{V}_t, l) \right\|_2 \right]. \tag{5}$$

This cumulative metric, $S_{l,c}^{(b)}$, naturally exhibits a stronger correlation with ultimate task success.

위와 같이 sensitivity를 기준으로 어떤 channel을 몇 bit로 양자화 할 지 결정



위 그림과 같이, single timestep과 full episode에 대한 sensitivity는 유사하기 때문에, full episode를 모두 누적하는 것보다 단순히 1 step에 대한 sensitivity를 계산

3. Overall Architecture

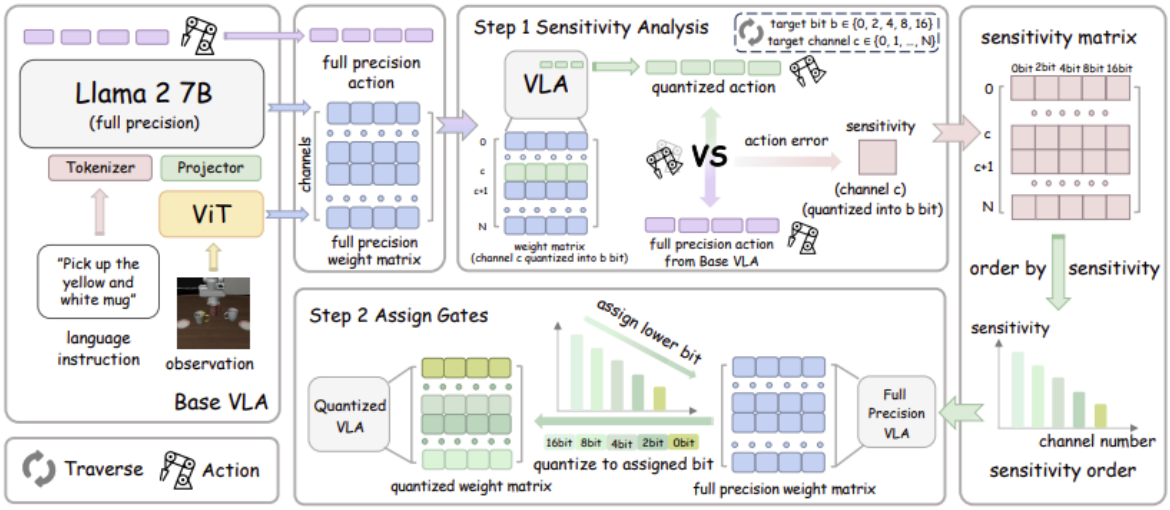


Figure 2: The pipeline of our **QVLA** framework consists of two steps: (i) In Step 1, we conduct a fine-grained action sensitivity analysis by systematically measuring and ranking the error induced by quantizing each channel to various bit-widths. (ii) In Step 2, an optimal bit-width is assigned to each channel using a greedy demotion algorithm, which iteratively prunes or lowers the precision of the least sensitive channels until the target bit budget is met.

QVLA의 과정은 매우 단순하다.

Step 1. 앞서 말한 channel 별 sensitivity를 계산한다.

Step 2. 그리고 sensitivity가 낮은 순서부터 quantization을 진행한다. (16,8,4,2,0 한단계씩)

Step 3. Model Size Threshold까지 이를 반복한다.

이때 sensitivity를 모두 계산하는 것은 복잡하다.

A rigorous theoretical foundation is indeed crucial for justifying our design choices. We have incorporated detailed theoretical analysis into the revised manuscript, providing the derivation for our first-order proxy used to efficiently estimate the action-space sensitivity.

The true sensitivity we aim to minimize is the action space Mean Squared Error (MSE) when quantizing layer l, channel c to bit-width b:

$$S_{l,c}^{(b)} := \mathbb{E}[\|\Delta A\|^2], \quad \Delta A = A(X_{l,c} + \Delta X_{l,c}^{(b)}) - A(X_{l,c}),$$

위의 MSE 식이 true sensitivity이지만, 각 채널 별 sensitivity를 구하기 위해 최종 action 까지 모든 layer와 action head를 지나는 full forward pass에 대한 cost가 크다.

이에 처음에는 1차 Taylor 전개로 sensitivity를 coarse하게 구하고, 이후 이를 바탕으로 coarse sensitivity가 높은 channel에 대해서만 full forward pass로 fine grained 한 sensitivity를 구한다.

First-order Taylor expansion으로 다음과 같이 간소화할 수 있다.

where $A(\cdot)$ is the mapping from the channel output to the final action, and $\Delta X_{l,c}^{(b)}$ is the quantization noise.

Assuming a local linearity for small perturbations, we apply a first-order Taylor expansion to approximate the action deviation:

$$\Delta A \approx J_{A,X_{l,c}} \Delta X_{l,c}^{(b)},$$

where $J_{A,X_{l,c}}$ is the Jacobian of the action A with respect to the channel output $X_{l,c}$. Substituting this approximation into $S_{l,c}^{(b)}$ yields:

$$S_{l,c}^{(b)} \approx \mathbb{E}[\|J_{A,X_{l,c}} \Delta X_{l,c}^{(b)}\|^2] = \mathbb{E}\Big[(\Delta X_{l,c}^{(b)})^\top \underbrace{(J_{A,X_{l,c}}^\top J_{A,X_{l,c}})}_{=:H_{l,c}} \Delta X_{l,c}^{(b)}\Big].$$

We then employ the common uniform quantization noise model, characterized by zero mean and an isotropic covariance structure:

$$\mathbb{E}[\Delta X_{l,c}^{(b)}] = 0, \quad \mathrm{Cov}(\Delta X_{l,c}^{(b)}) = (\sigma_{l,c}^{(b)})^2 I.$$

Under this assumption, the expected quadratic form simplifies via the matrix trace property $\mathbb{E}[\mathbf{x}^\top H \mathbf{x}] = \mathrm{Tr}(H\mathrm{Cov}(\mathbf{x}))$:

$$\mathbb{E}[(\Delta X_{l,c}^{(b)})^\top H_{l,c} \Delta X_{l,c}^{(b)}] = \mathrm{Tr}\big(H_{l,c}\,\mathrm{Cov}(\Delta X_{l,c}^{(b)})\big) = (\sigma_{l,c}^{(b)})^2 \mathrm{Tr}(H_{l,c}).$$

Since $\mathrm{Tr}(H_{l,c}) = \mathrm{Tr}(J_{A,X_{l,c}}^\top J_{A,X_{l,c}}) = \|J_{A,X_{l,c}}\|_F^2$ (the squared Frobenius norm of the Jacobian), we arrive at the final first-order approximation:

$$S_{l,c}^{(b)} \approx (\sigma_{l,c}^{(b)})^2 \cdot \|J_{A,X_{l,c}}\|_F^2.$$

This theoretical derivation rigorously justifies our efficient proxy, which factors the sensitivity into a predictable quantization error term $(\sigma_{l,c}^{(b)})^2)$ and a task-dependent local sensitivity gain $(\|J_{A,X_{l,c}}\|_F^2)$.

1. Action 변화량에 대한 MSE를 quantization error와 이에 따른 Jacobian으로 변환

2. FIM에 의해 Jacobian의 제곱을 Hessian으로 동치
3. 단일 channel 내에서 quantization error는 평균=0, 분산이 균일한 등방성 구조
4. 이에 xHx를 Trace 연산으로 변환 후 등방성에 따라, sensitivity는 (분산 제곱) x (Jacobian 절대값의 제곱)으로 변환된다.
5. 즉, Full Forward Pass 없이, 모델의 Jacobian과 현재 target channel만 확인하면 sensitivity를 바로 구할 수 있다.

즉, 최종 sensitivity는 아래와 같이, target channel의 출력 변화와 이에 대한 Jacobian의 곱으로 정리할 수 있다.

$$\|\Delta\mathcal{A}\| \approx \|J_{\mathcal{A},\mathbf{X}_{l,c}}\| \cdot \|\Delta\mathbf{X}_{l,c}\|,$$

이를 바탕으로 sensitivity를 반영하여, projector와 action head는 full precision을 유지하면서, language model과 vision encoder의 둔감한 channel에 대해 quantization 진행

perform a limited number of full forward passes, selectively targeting the most important channels to precisely calibrate their true sensitivity scores. This hybrid approach allows us to focus computational effort on preserving sensitive interfaces, such as the projector and action head, while enabling aggressive compression of less critical channels.